Machine Learning: Numerical Computation:
Overflow and Underflow – Gradient-based Optimization –
Constrained Optimization
Learning Algorithms: Capacity - Overfitting – Underfitting
Bayesian Classification
Supervised - unsupervised algorithms - Building
machine learning algorithm

Overflow and Underflow condition:

Aspect	Overflow	Underflow
Definition	Condition when an operation produces a value **greater than the maximum limit** that can be stored	Condition when an operation produces a value **less than the minimum limit** that can be stored
Occurs when	Exceeding upper bound of storage	Exceeding lower bound of storage
Common in	Arrays, stacks, queues, fixed-size memory, arithmetic operations	Stacks, queues, arithmetic operations
Example (Stack)	Pushing an element into a **full stack**	Popping an element from an **empty stack**
Example (Number)	Storing 130 in an 8-bit signed integer (range −128 to 127)	Result goes below −128 in an 8-bit signed integer
Effect	Data corruption or runtime error	Invalid operation or runtime error

Overfitting vs Underfitting (Machine Learning)

Aspect	Overfitting	Underfitting
Definition	Model learns **too much**, including noise in training data	Model learns **too little**, missing important patterns
Training error	Very low	High
Testing/Validation error	High	High
Model complexity	Too complex	Too simple
Generalization	Poor	Poor
Bias–Variance	Low bias, **high variance**	**High bias**, low variance
Typical cause	Too many features, deep models, small dataset	Insufficient features, simple model

Visual intuition
Underfitting → "Didn't learn enough"
Overfitting → "Learned everything, even noise"
Examples
Underfitting:
Using **linear regression** for a highly nonlinear dataset
Overfitting:
Very deep neural network trained on **small dataset**
How to fix

Problem	Solutions
Overfitting.	Regularization, dropout, more data, early stopping

Problem	Solutions
Underfitting	Increase model complexity, add features, train longer

One-line exam answers
Overflow: Exceeding the maximum storage limit
Underflow: Falling below the minimum storage limit
Overfitting: Model fits training data too closely
Underfitting: Model fails to capture data patterns

Bayesian Classification (Detailed Explanation)

1. What is Bayesian Classification?

Bayesian Classification is a probabilistic classification approach based on **Bayes' Theorem**, where we assign a class label to a data point by computing the **posterior probability** of each class and choosing the class with the highest probability.

Core idea:

"A data point belongs to the class that is most probable given the observed features."

2. Bayes' Theorem

$$P(C|X) = \frac{P(X|C)\ P(C)}{P(X)}$$

Where:
$P(C|X)$ → **Posterior probability** (what we want)
$P(X|C)$ → **Likelihood**
$P(C)$ → **Prior probability**
$P(X)$ → **Evidence** (normalizing constant)
For classification, $P(X)$ is same for all classes → can be ignored.

3. Decision Rule (MAP)

The classifier assigns class:

$$\hat{C} = argmax_C P(X|C)P(C)$$

This is called **Maximum A Posteriori (MAP)** decision rule.

4. Naive Bayesian Classifier

Why "Naive"?

Because it assumes **conditional independence** between features given the class:

$$P(X|C) = \prod_{i=1}^{n} P(x_i|C)$$

Even though this assumption is strong (often false), Naive Bayes works **surprisingly well**.

5. Types of Naive Bayes

Type	Used for
Gaussian NB	Continuous features
Multinomial NB	Count-based data (text, word frequency)

Type	Used for
Bernoulli NB	Binary features (0/1)

6. Algorithm (Naive Bayes)

Training Phase

Compute **prior probability**:

$$P(C_k) = \frac{No.\ of\ samples\ in\ C_k}{Total\ samples}$$

Estimate **likelihood**:
Gaussian NB:

$$P(x_i|C_k) = N(\mu_{ik}, \sigma_{ik}^2)$$

Prediction Phase

Compute posterior for each class
Select class with **maximum posterior probability**

7. Advantages & Limitations

Advantages

✔ Simple and fast
✔ Works well with high-dimensional data
✔ Requires small training data

Limitations

✘ Independence assumption
✘ Zero-frequency problem (handled by Laplace smoothing)

Simple Example of Bayesian Classification (ML)
Problem Statement
We want to classify whether a day is **Good for Playing Tennis** based on **Weather**.
Classes

$$C \in \{Play, Not\ Play\}$$

Feature

$$X = Weather \in \{Sunny, Rainy\}$$

Given Training Data

Day	Weather	Play Tennis
1	Sunny	Play
2	Sunny	Play
3	Sunny	Not Play
4	Rainy	Play
5	Rainy	Not Play

Step 1: Compute Prior Probabilities

$$P(Play) = \frac{3}{5} P(Not\ Play) = \frac{2}{5}$$

Step 2: Compute Likelihood
Weather = Sunny

$$P(Sunny|Play) = \frac{2}{3} P(Sunny|Not\ Play) = \frac{1}{2}$$

Step 3: Apply Bayes' Theorem
We want to classify a **Sunny day**.
Posterior for Play

$$P(Play|Sunny) \propto P(Sunny|Play)P(Play) = \frac{2}{3} \times \frac{3}{5} = \frac{2}{5}$$

Posterior for Not Play

$$P(Not\ Play|Sunny) \propto \frac{1}{2} \times \frac{2}{5} = \frac{1}{5}$$

Step 4: Decision Rule (MAP)

$$\frac{2}{5} > \frac{1}{5}$$

✅ **Classified as: Play Tennis**

Final Answer (One-line)
Using Bayesian classification, a sunny day is classified as **Play Tennis** because it has a higher posterior probability.

Same Example in Simple Python (No sklearn)

```
# Prior probabilities
P_play = 3/5
P_not_play = 2/5

# Likelihoods
P_sunny_given_play = 2/3
P_sunny_given_not_play = 1/2

# Posterior (ignoring denominator)
posterior_play = P_sunny_given_play * P_play
posterior_not_play = P_sunny_given_not_play * P_not_play

print("Posterior Play:", posterior_play)
print("Posterior Not Play:", posterior_not_play)

if posterior_play > posterior_not_play:
  print("Decision: Play Tennis")
else:
  print("Decision: Not Play")
```

Why this is called Naive Bayes?
If we had **multiple features** (Weather, Temperature, Humidity), we would assume:

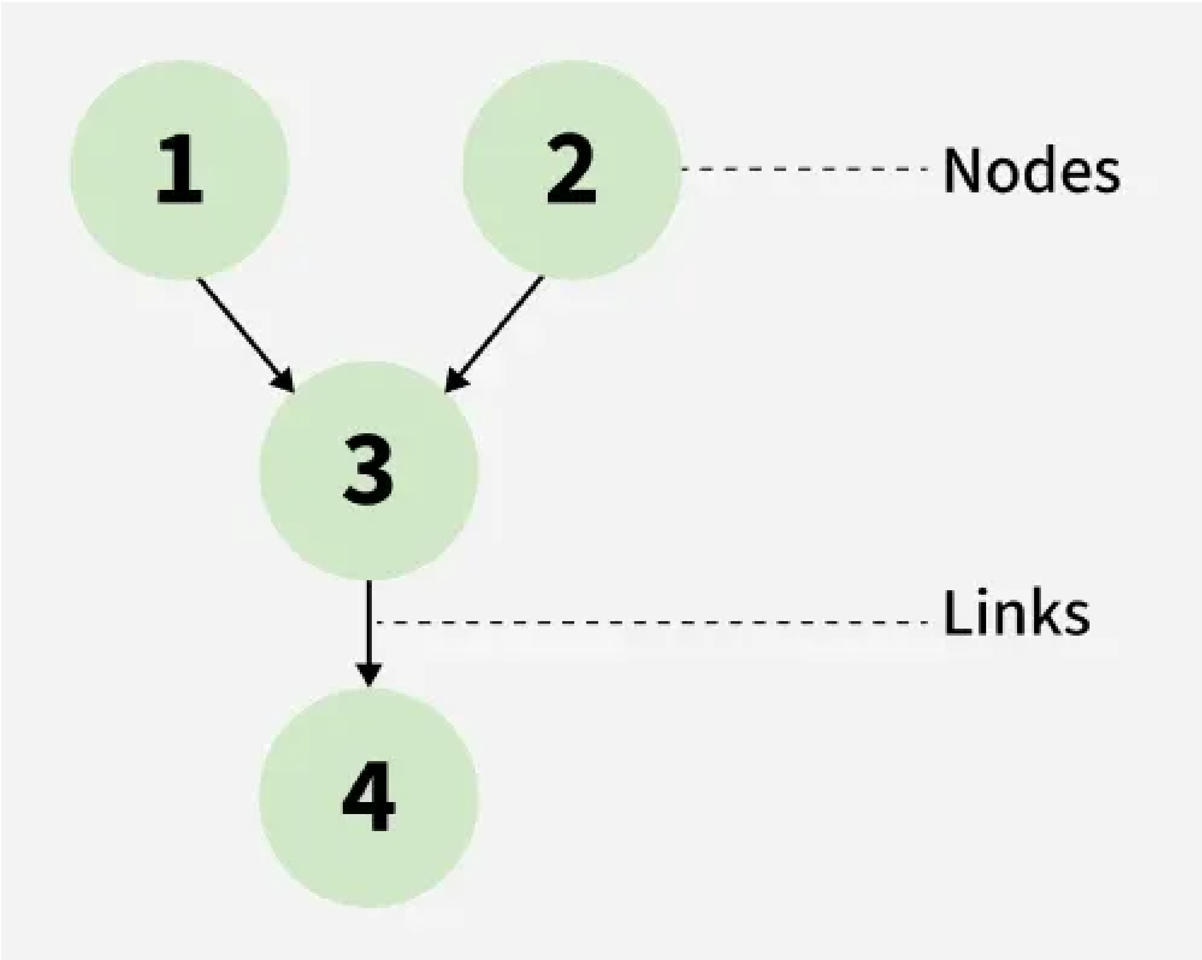
$$P(X_1, X_2, X_3|C) = P(X_1|C)P(X_2|C)P(X_3|C)$$

This **independence assumption** makes it *naive*.
Bayesian

Classification

Bayes Classification

Bayes Classification is a Supervised machine learning approach for classification. It works on a probabilistic method which uses Bayes Theorem to be implemented. It predicts the data point label or assigns the class on the basis of heuristic and statistical data. Let's dive deep into Bayes Classification and its Differences from Naive Bayes Classification.



Representation of Nodes and Links in a Bayesian Network

The above image denotes the relationship representation in a Bayes Classifier. The nodes represent features or variables in a Bayesian network. These can either be discrete or continuous. The links, on the other hand, denote the relationships between the nodes, along with a probability distribution over variables.

Key Features of Bayes Classifier

Probabilistic Model: Based on conditional probability using Bayes' Theorem.

Supervised Learning Approach: Requires labeled training data.
Classifies Using Prior and Likelihood: Combines prior class probabilities with data likelihood.
Model Interpretability: Easy to interpret due to mathematical formulation.
Scalability: Efficient for larger datasets.
Handles Continuous/Discrete Features: Works with both types depending on implementation.

Mathematical Representation

Bayes Theorem

Bayes' Theorem is a fundamental theorem in probability and machine learning that describes how to update the probability of an event when given new evidence. It is used as the basis of Bayes Classification.

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Where:
$P(C|X)$: Posterior probability of class C given data X
$P(X|C)$: Likelihood of data X given class C
$P(C)$: Prior probability of class C
$P(X)$: Marginal probability of data X

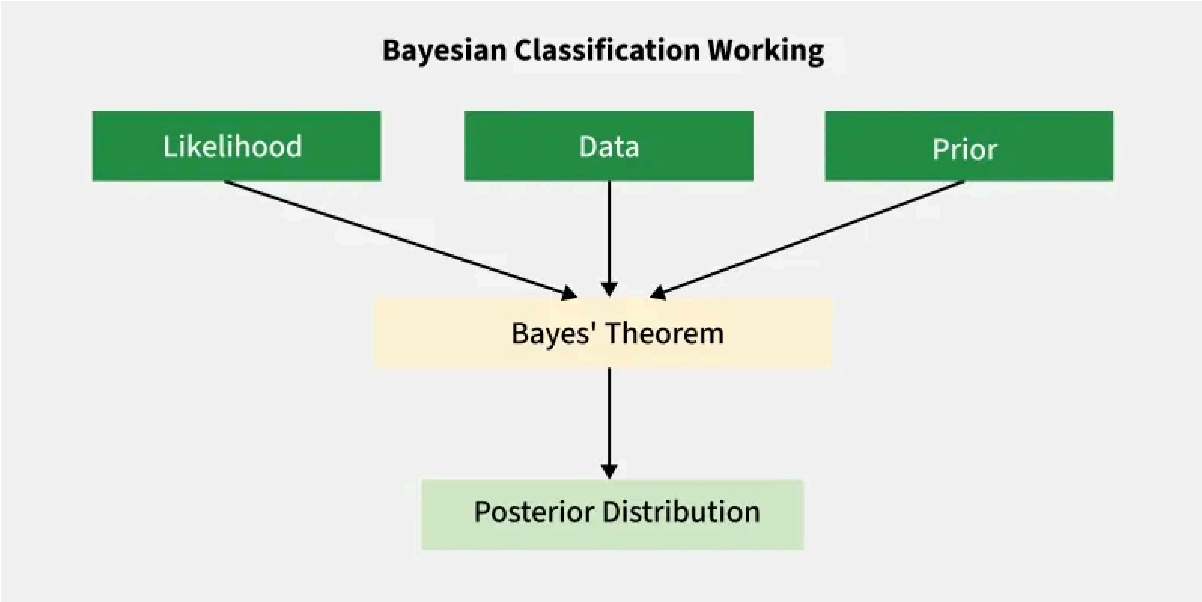
Assumptions in Bayes Classification

Well-defined prior probabilities.
Correct conditional probability distributions.
Independence of training and test samples.
Stationarity of features over time.
Classes are mutually exclusive and exhaustive.

Bayes Classification Workflow

Terminologies

Prior: Initial belief before evidence (e.g., % of spam emails). It is an Input to the Classifier.
Likelihood: Evidence corresponding to a particular given class (e.g., frequency of "free" in spam). It is an Input to the Classifier.
Posterior: Updated belief after evidence. It is an Output from the Classifier.
Evidence: Overall probability of observed features.



Bayesian Classification Working

The above image demonstrates that Likelihood, Data, and Prior probabilities are used as Input to the model. Bayes Theorem is used as the mathematical principle. The resultant is Posterior Distribution.

Steps Involved in Classification

Collect Data: From training set, extract features X and target C.
Estimate Priors: Calculate P(C) for each class.
Estimate Likelihoods: For every feature X, compute P(X|C).
Calculate Posterior: Use Bayes' Theorem to get P(C|X).
Predict Class: Assign the class with the highest posterior probability. It calculates the posterior probability for each class and assigns the class with the highest probability.

Bayes Classifier vs. Naive Bayes Classifier

Bayes Classifier is often misunderstood by its simplified version Naive Bayes Classifier. Here are some differences between them:

Factor	**Bayes Classifier**	**Naive Bayes Classifier**
Feature Dependency	Handles well	Assumes independence between features
Complexity	Complex	Simple and Fast
Interpretability	Clear Probabilities	Interpretable and Simplified logic

Factor	Bayes Classifier	Naive Bayes Classifier
Scalability	Less scalable	Highly scalable
Example	Bayesian Networks	Naive Bayes Classifier, Gaussian, Bernoulli, or Multinomial Classification

Why is it called a Bayes Classifier?

It's based on Bayes' Theorem, named after Thomas Bayes, an 18th-century statistician. The theorem helps update beliefs based on evidence, which is the core idea of classification here: updating class probability based on observed data.

Applications of Bayes Classification

Stock Market Prediction: Analyzes time-varying relationships and Financial indicators.

Fraud Detection and Credit Risk Modelling: Analyzes Fraud probability based on transaction patterns, contextual data, timestamps, etc.

Medical Diagnosis: Predicts the chances or probability of occurrence of a disease based on medical history.

Email Spam Detection and Phishing: Classifies emails and messages as "Spam" or "Not Spam" based on previous probabilities.

Advantages of Bayes Classifier

Handles small data well.
Can incorporate domain knowledge.
Probabilistic output allows threshold tuning.
Can model non-linear decision boundaries.
Adaptable to both discrete and continuous data.

Disadvantages of Bayes Classifier

Computationally expensive for high-dimensional data.
Complex to model feature dependencies.
Requires large data to estimate joint distributions.
Not scalable for many features or classes.